Benson Lin

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Education

University of California, Santa Barbara (UCSB) Expected June 2027
B.S. in Mechanical Engineering, GPA: 3.65 *Santa Barbara, CA*
Relevant Coursework: Adv. Viscous Flow (Grad), Thermosciences (I-II), Fluids (I-II), Control System Design, Mechatronics, Vibrations, Dynamical Systems, Structures, Applications of FEA, Machine Learning

Engineering and Research Experience

Gaucho Racing (UCSB Formula SAE Team) Sept 2023 – Present
Undertray & Composites Manufacturing Lead (2025 – Present), Aerodynamics Engineer (2023 – Present) *Santa Barbara, CA*
- Leading the design and development of the 2026 car's ground effects system, coordinating a small team from concept through simulation and manufacturing. Presented our aero architecture to industry judges at FSAE Michigan (30th overall).
- Applied iterative CFD and DOE testing to redesign side diffusers, achieving a 40% increase in theoretical downforce while maintaining aero balance and mitigating flow separation.
- Co-engineered key aerodynamic elements including the undertray, front/rear wings, and their mounting assemblies in SolidWorks. Optimized aerodynamics with Ansys Fluent CFD and ensured structural integrity through hand calculations and FEA.
- Managed the manufacturing workflow from CAD to fabricated components utilizing carbon fiber/fiberglass wet layups, foam/honeycomb cores, and vacuum bagging, ensuring precise integration with other vehicle systems.

UCSB Undergraduate Research Project (URCA) August 2025 – May 2026
Design and Analysis Lead *Santa Barbara, CA*
- Led a multidisciplinary team of 4 to secure a $1,200 UCSB URCA grant to develop Formula SAE vehicle sidepods for improved cooling system performance.
- Established an Ansys Fluent CFD pipeline coupling external aerodynamics with internal radiator duct flow across four configurations, predicting a 46% increase in radiator mass flow rate. Results informed lap simulations by the cooling subteam, estimating a 63% improvement in thermal endurance.

Bio-Inspired Fluid Flow Lab Sept 2025 – June 2026
Undergraduate Researcher *Santa Barbara, CA*
- Investigated Marangoni flow phenomena by designing and running experiments on decane droplet spreading across aqueous surfactant baths, characterizing the transition between Laplace- and Marangoni-dominated regimes.
- Conducted a parameter sweep across three vessel diameters and two surfactant concentrations using high-speed videography and ImageJ/MATLAB analysis, finding that smaller vessels cause premature plateauing from wall pinning.
- Validated the measurement method with power-law fitting that matched an established lab baseline within ∼4%. Designed & 3D-printed custom mounting structures to improve data collection precision.

Projects

Disney Imagineering Senior Capstone: Sensor-Fusion Robotic Arm Fall 2026 – Present
- Part of a 5-person team ($6K budget) developing a robotic arm with a multi-sensor package to detect its environment and human interactions through sensor fusion.

Computer Vision-Guided Automated Hydration System – Junior Capstone Spring 2026
- Led a four-person team in building a hands-free hydration system for helmeted athletes on the sidelines, tracking a person's mouth and delivering water under a $200 budget.
- Designed a vision-in-the-loop pan-tilt control system, streaming Python/MediaPipe face-tracking data to an Arduino running a PD controller. Added derivative control and a deadzone to fix jitter and hunting near the target.
- Developed and calibrated facial landmark-based distance estimation using physical measurements. Implemented distance-dependent water trajectory compensation with empirical corrections to improve aiming across varying distances.
- Sized servos using moment of inertia calculations (2.35 torque factor of safety) and resolved pump-induced control signal interference with a flyback diode, achieving reliable water delivery at 1.21 m against a 1.5 m design target.

Skills

Design & CAD:	SolidWorks, Fusion360, Technical Drawings, Rapid Prototyping, GD&T
Simulation & Analysis:	Ansys Workbench, Ansys Fluent (CFD, DOE), COMSOL, Finite Element Analysis
Fabrication & Machining:	CNC Mill, Manual Lathe, Bandsaw, 3D Printing, Composites (Wet Layup, Bagging, Cores)
Programming:	MATLAB, Python (NumPy, PyTorch, Pyomo), Java, Arduino, LabVIEW
Other:	Microsoft Office Suite, LaTeX; English (Fluent), Chinese (Fluent)

Activities and Interests

UCSB Wind Ensemble & Orchestra – *Principal Clarinetist* *2023 – Present*